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Saveetha Institute of Medical And Technical Sciences  
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

# **Smart Textile Manufacturing Quality Inspection and Production Optimization Platform**

## **AGILE SPRINT PLANNING & USER STORY WORKSHOP**

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# 1. Project Vision and Stakeholders

## 1.1 Project Vision

The **Smart Textile Manufacturing Quality Inspection and Production Optimization Platform** aims to transform traditional textile manufacturing into a smart, automated, and data-driven production environment. The platform integrates Artificial Intelligence (AI), Computer Vision, Internet of Things (IoT), and Data Analytics to automate fabric defect detection, monitor production processes in real time, and optimize manufacturing efficiency.

The system replaces manual inspection methods by automatically detecting fabric defects such as weaving faults, stitching errors, dyeing inconsistencies, stains, and surface damages during production. It also provides real-time monitoring of machine performance, production output, and operational efficiency through interactive dashboards. By combining quality inspection with production analytics, the platform enables manufacturers to reduce material wastage, improve product quality, optimize machine utilization, and support faster decision-making.

The vision of the project is to build an intelligent textile manufacturing platform that enhances productivity, minimizes production costs, improves customer satisfaction, and supports continuous process improvement through automation and data-driven insights.

## 1.2 Project Objectives

The Smart Textile Manufacturing Quality Inspection and Production Optimization Platform has the following objectives:

- To develop an AI-based fabric defect detection system capable of identifying weaving, dyeing, stitching, and surface defects automatically.
- To implement a real-time production monitoring dashboard that continuously displays machine status, production output, machine utilization, downtime, and operational performance.
- To design a defect classification module that categorizes detected defects based on their type, severity, and location on the fabric.
- To implement a root-cause analysis system that correlates defect occurrences with production machines, operators, work shifts, and raw material batches.

- To develop a production optimization engine that recommends machine speed adjustments and production scheduling improvements using historical manufacturing data.
- To generate automated notifications whenever defect rates exceed predefined quality thresholds, enabling supervisors to take immediate corrective actions.

### 1.3 Stakeholders

The successful implementation of the platform requires collaboration among multiple stakeholders. Each stakeholder has a specific responsibility in ensuring smooth production, maintaining quality standards, and improving manufacturing efficiency.

| STAKEHOLDER                       | ROLE AND RESPONSIBILITY                                                                                                      |
|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| <b>Factory Owner</b>              | Reviews overall production performance, monitors business growth, and makes strategic decisions based on production reports. |
| <b>Production Manager</b>         | Supervises manufacturing operations, production planning, machine utilization, and workforce management.                     |
| <b>Quality Inspector</b>          | Monitors fabric quality, verifies detected defects, and ensures products meet quality standards before delivery.             |
| <b>Machine Operator</b>           | Operates textile machinery, monitors machine conditions, and reports operational issues.                                     |
| <b>Maintenance Engineer</b>       | Performs preventive and corrective maintenance of textile machines and resolves equipment failures.                          |
| <b>System Administrator</b>       | Manages user accounts, system security, database maintenance, and software configuration.                                    |
| <b>AI Defect Detection System</b> | Automatically detects, classifies, and records defects using computer vision and machine learning algorithms.                |
| <b>Production Supervisor</b>      | Monitors production progress, receives quality alerts, and coordinates corrective actions on the production floor.           |
| <b>Customers</b>                  | Receive high-quality textile products with minimal manufacturing defects and timely order delivery.                          |

## 1.4 Stakeholder Responsibilities Summary

| Stakeholder           | Primary Goal                                   |
|-----------------------|------------------------------------------------|
| Factory Owner         | Improve business performance and profitability |
| Production Manager    | Increase production efficiency                 |
| Quality Inspector     | Ensure product quality                         |
| Production Supervisor | Reduce production defects                      |
| Machine Operator      | Maintain smooth machine operation              |
| Maintenance Engineer  | Reduce machine downtime                        |
| System Administrator  | Ensure system availability and security        |
| AI System             | Detect and classify defects automatically      |
| Customer              | Receive high-quality textile products          |

## 2.Epics and User Stories

### 2.1 Overview

The Smart Textile Manufacturing Quality Inspection and Production Optimization Platform is divided into multiple Epics, each representing a major functional area of the system. Every Epic consists of several User Stories that describe the system requirements from the perspective of different stakeholders. These User Stories follow the Agile format: "As a <user>, I want <feature>, so that <benefit>." This approach helps the development team understand user requirements, prioritize features, and deliver the project incrementally through multiple sprints.

### Epic 1 – User Authentication and User Management

#### Description:

This Epic focuses on providing secure authentication and authorization mechanisms for all users of the platform. It ensures that only authorized personnel can access the system and perform their respective roles based on assigned permissions.

### **US1.1 – Create User Accounts**

As a System Administrator, I want to create user accounts so that authorized users can access the platform.

### **US1.2 – Secure User Login**

As a Factory Employee, I want secure login so that I can access my assigned features.

### **US1.3 – Role-Based Access Control**

As a System Administrator, I want role-based permissions so that users access only authorized modules.

## **Epic 2 – Fabric Defect Detection**

### **Description:**

This Epic automates the inspection process by using Artificial Intelligence and Computer Vision techniques to detect fabric defects accurately and efficiently during production.

### **US2.1 – Automatic Fabric Defect Detection**

As a Quality Inspector, I want automatic defect detection so that manual inspection time is reduced.

### **US2.2 – Defect Location Identification**

As a Quality Inspector, I want defects highlighted on images so that I can identify their location.

### **US2.3 – Real-Time Defect Display**

As a Production Supervisor, I want immediate defect display so that corrective actions can be taken.

## **Epic 3 – Defect Classification**

### **Description:**

This Epic classifies detected defects according to their type, severity, and location to assist quality inspectors and production managers in taking appropriate corrective actions.

### **US3.1 – Defect Type Classification**

As a Quality Inspector, I want defect type classification so that corrective actions are easier.

### **US3.2 – Defect Severity Classification**

As a Production Manager, I want severity classification so that priorities are set.

### **US3.3 – Defect Location Mapping**

As a Quality Inspector, I want defect location mapping so repairs are easier.

## **Epic 4 – Production Monitoring Dashboard**

### **Description:**

This Epic provides a centralized dashboard that enables real-time monitoring of machine performance, production output, downtime, and operational efficiency.

### **US4.1 – Real-Time Machine Performance Monitoring**

As a Production Manager, I want real-time monitoring so that efficiency improves.

### **US4.2 – Production Output Monitoring**

As a Supervisor, I want production output monitoring so targets are achieved.

### **US4.3 – Factory Performance Dashboard**

As a Factory Owner, I want dashboards so that overall performance is visible.

## **Epic 5 – Root Cause Analysis**

### **Description:**

This Epic helps identify the root causes of recurring defects by analyzing production data, machine performance, work shifts, and raw material batches.

### **US5.1 – Machine Defect Analysis**

As a Maintenance Engineer, I want machine defect analysis so preventive maintenance can be scheduled.

### **US5.2 – Shift-Wise Defect Analysis**

As a Production Manager, I want shift-wise analysis so operational issues are identified.

## **Epic 6 – Alerts and Notifications**

### **Description:**

This Epic provides automatic notifications whenever abnormal production conditions or excessive defect rates are detected.

### **US6.1 – High Defect Rate Alert**

As a Supervisor, I want alerts for high defect rates so corrective action is immediate.

### **US6.2 – Machine Failure Notification**

As a Maintenance Engineer, I want machine failure notifications so downtime is minimized.

## **Epic 7 – Reporting and Analytics**

### **Description:**

This Epic generates centralized production and quality reports to support performance evaluation and business decision-making.

### **US7.1 – Production Report Generation**

As a Factory Owner, I want production reports so that I can monitor performance.

### **US7.2 – Defect Analysis Report**

As a Quality Inspector, I want defect reports so quality trends can be reviewed.

### **US7.3 – Production Performance Comparison**

As a Production Manager, I want performance comparison so productivity improvements can be evaluated.

## **Epic 8 – Production Optimization**

### **Description:**

This Epic focuses on improving manufacturing efficiency by analyzing production data and recommending optimized machine settings and production schedules.

### **US8.1 – Machine Speed Optimization**

As a Production Manager, I want optimized machine speed recommendations so efficiency improves.

### **US8.2 – Production Scheduling Recommendation**

As a Factory Owner, I want scheduling recommendations so utilization improves.

### **US8.3 – Production Bottleneck Reduction**

As a Supervisor, I want bottleneck reduction recommendations so orders finish on time.

## **3.Acceptance Criteria**

### **3.1 Overview**

Acceptance Criteria define the specific conditions that must be met before a User Story is considered complete. They ensure that the developed feature satisfies user requirements, meets quality standards, and functions as expected. Acceptance Criteria also help developers, testers, and stakeholders clearly understand the expected outcome of each User Story.

### **3.2 Acceptance Criteria for User Stories**

#### **US1.1 – Create User Accounts**

##### **Acceptance Criteria**

- The System Administrator can create a new user account.
- All mandatory fields must be completed.
- Each user account must have a unique username.
- The system assigns the selected user role successfully.
- A confirmation message is displayed after successful account creation.

#### **US2.1 – Automatic Fabric Defect Detection**

##### **Acceptance Criteria**

- The system accepts fabric images from the inspection camera.
- AI automatically analyzes the uploaded image.
- Fabric defects are accurately detected.
- Detected defects are highlighted on the image.
- Inspection results are displayed within a few seconds.



### **US3.1 – Defect Type Classification**

#### **Acceptance Criteria**

- The system classifies each detected defect.
- Defect type is displayed correctly.
- Severity level is assigned automatically.
- Classification results are stored in the database.
- Users can review previous classifications.

### **US4.1 – Real-Time Machine Performance Monitoring**

#### **Acceptance Criteria**

- Machine status is updated in real time.
- Production count is displayed correctly.
- Machine downtime is recorded automatically.
- Dashboard refreshes without manual intervention.
- Production statistics are visible to authorized users.

### **US5.1 – Machine Defect Analysis**

#### **Acceptance Criteria**

- The system identifies machines with recurring defects.
- Historical defect data is analyzed.
- Root causes are displayed.
- Maintenance recommendations are generated.
- Analysis reports can be viewed by managers.

### **US6.1 – High Defect Rate Alert**

#### **Acceptance Criteria**

- The system continuously monitors defect rates.
- Alerts are generated when the threshold is exceeded.
- Notifications are sent immediately.
- Alert details include machine ID and defect count.
- Alert history is stored for future reference.

## **US7.1 – Production Report Generation**

### **Acceptance Criteria**

- Users can generate production reports.
- Reports include production statistics.
- Reports include defect summaries.
- Reports can be downloaded in PDF format.
- Only authorized users can access reports.

## **US8.1 – Machine Speed Optimization**

### **Acceptance Criteria**

- The system analyzes production data.
- Machine speed recommendations are generated.
- Recommendations improve production efficiency.
- Managers can review optimization suggestions.
- Optimization history is stored for future analysis.

### **3.3 Summary of Acceptance Criteria**

| <b>User Story</b> | <b>Acceptance Criteria Summary</b>                   |
|-------------------|------------------------------------------------------|
| <b>US1.1</b>      | User account created successfully with valid details |
| <b>US2.1</b>      | AI detects and highlights fabric defects             |
| <b>US3.1</b>      | Defects are classified correctly                     |
| <b>US4.1</b>      | Dashboard displays real-time production data         |
| <b>US5.1</b>      | Root causes of defects are identified                |
| <b>US6.1</b>      | Alerts are generated for high defect rates           |
| <b>US7.1</b>      | Production reports are generated and downloaded      |
| <b>US8.1</b>      | Optimization recommendations are provided            |

## **4. Product Backlog Prioritization**

### **4.1 Overview**

The Product Backlog is a prioritized list of all features, enhancements, and functionalities required to successfully develop the Smart Textile Manufacturing Quality Inspection and Production Optimization Platform. The backlog helps the development team focus on delivering the most valuable features first. Priorities are assigned based on business value, user importance, and technical dependency.

### **4.2 Product Backlog Priority Levels**

#### **High Priority**

These features are essential for the first release because they provide the core functionality of the system, including secure authentication, AI-based defect detection, production monitoring, and automatic alerts.

- User Authentication
- Secure Login
- Role-Based Access
- Automatic Fabric Defect Detection
- Defect Location Identification
- Real-Time Defect Display
- Machine Performance Monitoring
- High Defect Rate Alerts
- Machine Failure Notifications

#### **Medium Priority**

These features enhance the system by providing additional analytics, classification, reporting, and optimization capabilities.

- Defect Classification
- Severity Classification
- Location Mapping
- Production Output Monitoring
- Factory Dashboard
- Machine Defect Analysis

- Production Reports
- Defect Reports
- Machine Speed Optimization
- Production Scheduling
- Bottleneck Reduction

### **Low Priority**

These features improve long-term analysis and decision-making but are not mandatory for the first software release.

- Shift-Wise Defect Analysis
- Production Performance Comparison

## **4.3 Product Backlog Summary**

| <b>Priority</b> | <b>Number of User Stories</b> |
|-----------------|-------------------------------|
| <b>High</b>     | 9                             |
| <b>Medium</b>   | 11                            |
| <b>Low</b>      | 2                             |
| <b>Total</b>    | 22 User Stories               |

# **5.Sprint Backlog Planning**

## **5.1 Overview**

The Sprint Backlog is a subset of the Product Backlog selected for implementation during a sprint. It contains the user stories, development tasks, estimated effort, and team members responsible for completing the work. The goal of Sprint 1 is to deliver the core functionalities of the Smart Textile Manufacturing Quality Inspection and Production Optimization Platform.

## **5.2 Sprint Goal**

The primary goal of Sprint 1 is to develop the core modules of the Smart Textile Manufacturing Quality Inspection and Production Optimization Platform, including secure user authentication, AI-based fabric defect detection, real-time production monitoring, and

alert notifications. These features provide the foundation for the remaining functionalities in future sprints.

### 5.3 Sprint Duration

| Sprint   | Duration          |
|----------|-------------------|
| Sprint 1 | 2 Weeks (14 Days) |

### 5.4 Definition of Done (DoD)

The primary goal of Sprint 1 is to develop the core modules of the Smart Textile Manufacturing Quality Inspection and Production Optimization Platform, including secure user authentication, AI-based fabric defect detection, real-time production monitoring, and alert notifications. These features provide the foundation for the remaining functionalities in future sprints.

### 5.5 Sprint Deliverables

At the end of Sprint 1, the following deliverables will be completed:

- User Registration Module
- Secure Login System
- Role-Based Access Control
- AI-Based Fabric Defect Detection Module
- Real-Time Production Monitoring Dashboard

## 6. Story Point Estimation

### 6.1 Overview

Story Point Estimation is an Agile technique used to estimate the relative effort required to implement each User Story. Instead of measuring the work in hours, Story Points consider the complexity, effort, and uncertainty involved in completing a task. In this project, the **Fibonacci sequence (1, 2, 3, 5, 8, 13)** is used for estimation because it is a widely accepted Agile estimation method.

## 6.2 Story Point Estimation Scale

| Story Points | Complexity     | Description                                                      |
|--------------|----------------|------------------------------------------------------------------|
| 1            | Very Low       | Very simple task with minimal effort                             |
| 2            | Low            | Small enhancement or minor functionality                         |
| 3            | Medium         | Standard feature with moderate effort                            |
| 5            | High           | Complex feature requiring multiple tasks                         |
| 8            | Very High      | Advanced functionality with significant development effort       |
| 13           | Extremely High | Very complex feature requiring extensive development and testing |

## 6.3 Story Point Estimation Table

| User Story ID | User Story                               | Story Points | Complexity |
|---------------|------------------------------------------|--------------|------------|
| US1.1         | Create User Accounts                     | 3            | Medium     |
| US1.2         | Secure User Login                        | 2            | Low        |
| US1.3         | Role-Based Access Control                | 5            | High       |
| US2.1         | Automatic Fabric Defect Detection        | 8            | Very High  |
| US2.2         | Defect Location Identification           | 5            | High       |
| US2.3         | Real-Time Defect Display                 | 3            | Medium     |
| US3.1         | Defect Type Classification               | 5            | High       |
| US3.2         | Defect Severity Classification           | 3            | Medium     |
| US3.3         | Defect Location Mapping                  | 3            | Medium     |
| US4.1         | Real-Time Machine Performance Monitoring | 5            | High       |

|              |                                      |   |           |
|--------------|--------------------------------------|---|-----------|
| <b>US4.2</b> | Production Output Monitoring         | 3 | Medium    |
| <b>US4.3</b> | Factory Performance Dashboard        | 5 | High      |
| <b>US5.1</b> | Machine Defect Analysis              | 5 | High      |
| <b>US5.2</b> | Shift-Wise Defect Analysis           | 3 | Medium    |
| <b>US6.1</b> | High Defect Rate Alert               | 5 | High      |
| <b>US6.2</b> | Machine Failure Notification         | 5 | High      |
| <b>US7.1</b> | Production Report Generation         | 3 | Medium    |
| <b>US7.2</b> | Defect Analysis Report               | 3 | Medium    |
| <b>US7.3</b> | Production Performance Comparison    | 2 | Low       |
| <b>US8.1</b> | Machine Speed Optimization           | 8 | Very High |
| <b>US8.2</b> | Production Scheduling Recommendation | 5 | High      |
| <b>US8.3</b> | Production Bottleneck Reduction      | 5 | High      |

#### 6.4 Total Story Points

| <b>Sprint</b>                    | <b>Total User Stories</b> | <b>Total Story Points</b> |
|----------------------------------|---------------------------|---------------------------|
| <b>Sprint 1</b>                  | 8                         | 36                        |
| <b>Remaining Product Backlog</b> | 14                        | 68                        |
| <b>Overall Project</b>           | 22                        | 104                       |

#### 6.5 Estimation Justification

The Story Point values were assigned based on the complexity, development effort, technical challenges, and testing requirements of each User Story.

- **Low-complexity tasks (2–3 points)** include user login, account creation, and report generation, as they require standard development effort.

- **Medium-complexity tasks (3–5 points)** include dashboards, defect classification, machine monitoring, and alert notifications because they involve multiple components and integrations.
- **High-complexity tasks (8 points)** include AI-based fabric defect detection and machine speed optimization, as they require machine learning algorithms, image processing, data analysis, and extensive testing.

## 6.6 Benefits of Story Point Estimation

- Helps estimate development effort accurately.
- Improves sprint planning and workload distribution.
- Enables better resource allocation.
- Reduces project risks by identifying complex features early.
- Supports predictable sprint delivery.
- Improves communication among Agile team members.

## 6.7 Summary

Story Point Estimation provides a structured approach for evaluating the effort required to implement each User Story. By using the Fibonacci estimation technique, the development team can effectively plan sprints, balance workloads, and deliver high-quality software within the planned schedule.

## 7.Sprint Goal and Expected Deliverables

The Smart Textile Manufacturing Quality Inspection and Production Optimization Platform provides an effective solution for improving quality control and production management in textile industries. By integrating Artificial Intelligence, Computer Vision, IoT, and Data Analytics, the system automates defect detection, monitors production in real time, and supports data-driven decision-making. Compared to the existing manual process, the proposed platform improves inspection accuracy, reduces material wastage, optimizes machine utilization, and enhances overall manufacturing efficiency. The use of Agile Scrum as the development methodology ensures flexibility, scalability, and easier maintenance, making the system suitable for future expansion. Overall, the proposed solution helps textile manufacturers achieve higher productivity, better product quality, and increased customer satisfaction.



## 7.1 Overview

The Sprint Goal defines the primary objective that the Agile team aims to achieve during the sprint. Expected Deliverables are the functional modules, documents, and software components completed at the end of the sprint. These deliverables provide measurable progress toward the successful development of the Smart Textile Manufacturing Quality Inspection and Production Optimization Platform.

## 7.2 Sprint Goal

The goal of **Sprint 1** is to develop the core foundation of the Smart Textile Manufacturing Quality Inspection and Production Optimization Platform by implementing secure user authentication, AI-based fabric defect detection, real-time production monitoring, and an automated alert system. Successfully completing these features will establish a functional prototype that supports future enhancements and additional sprint development.

## 7.3 Sprint Objectives

The primary objectives of Sprint 1 are:

- Develop a secure user authentication and authorization system.
- Implement AI-based fabric defect detection.
- Display detected defects in real time.
- Develop a production monitoring dashboard.
- Generate automatic alerts for defects and machine failures.

## 7.4 Expected Deliverables

At the end of Sprint 1, the following deliverables will be completed:

| Deliverable ID | Deliverable                             | Description                                                        |
|----------------|-----------------------------------------|--------------------------------------------------------------------|
| D1             | User Authentication Module              | User registration, login, and role-based access control.           |
| D2             | AI-Based Fabric Defect Detection Module | Automatic detection and identification of fabric defects using AI. |

|    |                                 |                                                                    |
|----|---------------------------------|--------------------------------------------------------------------|
| D3 | Production Monitoring Dashboard | Dashboard displaying real-time production and machine status.      |
| D4 | Alert and Notification System   | Automatic alerts for high defect rates and machine failures.       |
| D5 | Database Design                 | Database for storing users, production data, defects, and reports. |
| D6 | System Integration              | Integration of all developed modules into a working prototype.     |
| D7 | Testing Report                  | Unit testing, integration testing, and bug-fixing report.          |

## 7.5 Sprint Outcome

At the completion of Sprint 1, the development team will deliver a working prototype of the Smart Textile Manufacturing Quality Inspection and Production Optimization Platform. The prototype will include secure authentication, AI-based defect detection, production monitoring, and alert generation. These completed features will serve as the foundation for future sprints, where advanced analytics, reporting, optimization, and additional functionalities will be implemented.

## 7.6 Benefits of Sprint Deliverables

The successful completion of Sprint 1 provides the following benefits:

- Establishes a strong foundation for future development.
- Improves fabric quality through AI-powered defect detection.
- Reduces manual inspection time and production errors.
- Enables real-time monitoring of production activities.
- Enhances decision-making with timely alerts and notifications.
- Supports efficient Agile development through incremental feature delivery.

## **7.7 Sprint Review Summary**

At the Sprint Review meeting, stakeholders will verify that:

- All selected User Stories have been completed.
- Acceptance Criteria are satisfied.
- The system functions as expected.
- AI detects and displays fabric defects accurately.
- The dashboard shows real-time production information.
- Alerts are generated successfully.
- Feedback is collected for the next sprint.

## **7.8 Conclusion**

The Agile Sprint Planning process provides a structured approach to developing the Smart Textile Manufacturing Quality Inspection and Production Optimization Platform. By defining Epics, User Stories, Acceptance Criteria, Product Backlog, Sprint Backlog, Story Point Estimation, Sprint Goals, and Expected Deliverables, the development team can effectively manage project requirements, prioritize tasks, and deliver high-quality software incrementally. This approach improves collaboration, enhances productivity, and ensures the successful implementation of a smart, AI-driven textile manufacturing solution.